

Gábor Hosu

contact@hgabor.com | +40 742 526 970 | hgabor.com | github.com/gabor-hosu | linkedin.com/in/gabor-hosu

I am a university student and Junior Data Scientist with a strong mathematical foundation and hands-on backend development experience. I am passionate about designing and building machine learning systems, especially in natural language processing, information retrieval, and applied data science.

Honours and Publications

- Á. Kránics, **G. Hosu**, et al., "Multimodal RAG for Context-Aware Chatbots", Oct 2025
IEEE SISY 23rd | [DOI], [Preprint]
- 2nd Prize** at 28th Transylvanian Students' Scientific Conference May 2025

Skills

Languages: Python, SQL (MS SQL)

ML/AI: PyTorch, Transformers, LangChain, Sentence-Transformers, Hugging Face

Data: NumPy, Pandas, Milvus **Backend/Infra:** FastAPI, SQLAlchemy, Docker, Azure, GitLab CI/CD

Other: Git, Linux **Familiar with:** OpenCL, Java, Spring, React, C++

Experience

- Junior Data Scientist | Qwertee Technology** Aug 2026 – Present
- Software Engineering Intern | Codespring** Jul 2024 – Sept 2024
- Designed backend for a **[RAG chatbot]**: document ingestion, embedding, and retrieval pipeline.
 - Built and deployed REST APIs (auth, file management) via Docker and GitLab CI/CD to Azure.
- Fullstack mentorprogram | Codespring** Oct 2023 - May 2024
- Pre-internship training program covering fullstack development with Java, Spring, and React.

Projects

- Legal Document Form-Filling Prediction** Apr 2026 – May 2026
- Designed a lightweight ML model for legal document field-type prediction using text and layout context, targeting low-data settings where OCR-based approaches fail.
 - Built and annotated a dataset of 198 legal PDFs (19 classes); achieved **75% macro F1** (5-fold CV) with a **hierarchical coarse-to-fine classification** approach.
- Height Map Estimation From Normal Maps** Mar 2026 – May 2026
- Built GPU-based Poisson surface reconstruction pipeline from normal maps; derived and implemented **finite-difference** and **FFT solvers** from scratch.
 - Benchmarked iterative (**Jacobi**, **Gauss-Seidel**) and **spectral methods** in **OpenCL** + Python with Docker; evaluated scalability and accuracy on large-scale image grids.
- Embedding-Based Retrieval Paper Reproduction** Nov 2025 – Jan 2026
[Portfolio] | [GitHub] | [Paper] | [HF (Models & Data)]
- Reproduced a **2025 Google DeepMind paper** on geometric limits of single-vector embedding retrieval under GPU constraints.
 - Built and evaluated synthetic IR pipelines (2K–50K docs), benchmarked quantized LLMs, tuned hyperparameters; presented findings at a research group seminar.

Education

- Babeş-Bolyai University**, MSc, Data Analysis and Modelling Oct 2025 – Present
Relevant coursework: NLP, Information Retrieval, Machine Learning, GPU Programming, Swarm Intelligence, Robotics
- Babeş-Bolyai University**, BSc, Mathematics and Computer Science 2022 – 2025
GPA: 9.58/10 | Thesis: "A file-centric chatbot based on the RAG method"
- Languages:** English (Upper-Intermediate) Romanian (Intermediate) Hungarian (Native)